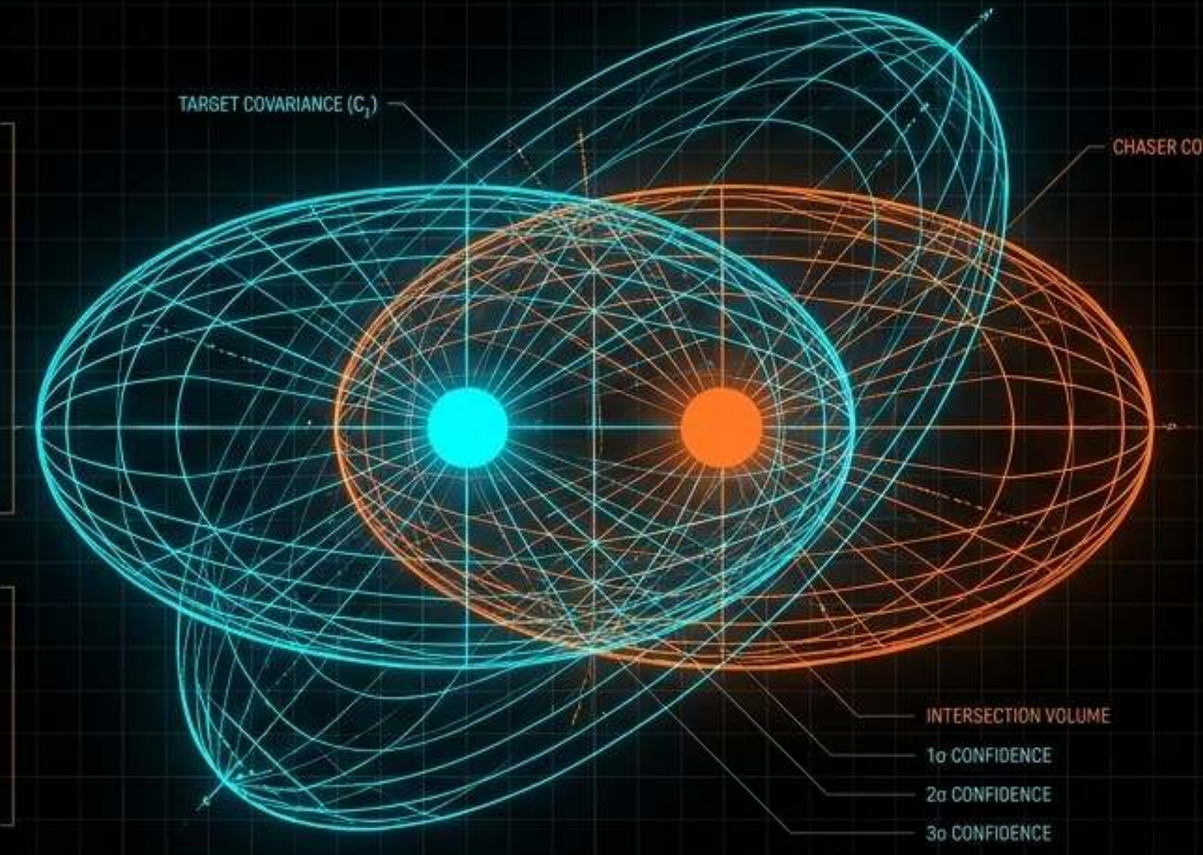
LAYER 03 // STATISTICAL ANATOMY OF THE CDM

The Architecture of Uncertainty

A mathematical deep-dive into the 48 statistical columns driving collision risk, covariance, and the Data Dilution paradox.

$$C_T = \begin{bmatrix} \sigma^2_x & \sigma_{xy} & \sigma_{xz} & \sigma_x V_x & \sigma_x V_y & \sigma_x V_z \\ \sigma_{yx} & \sigma^2_y & \sigma_{yz} & \sigma_y V_x & \sigma_y V_y & \sigma_y V_z \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



$$C_T = \begin{bmatrix} \sigma^2_x & \sigma_{xy} & \sigma_x V_x & \sigma_x V_y & \sigma_x V_z \\ \sigma_{yx} & \sigma^2_y & \sigma_y V_x & \sigma_y V_y & \sigma_y V_z \\ \sigma^2_y & \sigma_{yz} & \sigma_y V_x & \sigma_y V_y & \sigma_y V_z \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$J_T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \cdots \end{bmatrix}$$

$$C_C = \begin{bmatrix} \sigma^2_x & \sigma_{xy} & \sigma_{xz} & \sigma_x V_y & \sigma_x V_z \\ \sigma_{yx} & \sigma^2_y & \sigma_{yz} & \sigma_y V_y & \sigma_y V_z \\ \sigma^2_x & \sigma^2_y & \sigma_{yz} & \sigma_y V_x & \sigma_y V_z \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \sigma_{yx} & \sigma^2_y & \sigma_y V_x & \sigma_y V_y & \sigma_y V_z \end{bmatrix}$$

$$J_C = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \cdots \end{bmatrix}$$

CDM PARAMETERS

RELATIVE VELOCITY [km/s]:	13.87 (Cyan/Orange)
MISS DISTANCE [m]:	45.2 (Cyan/Orange)
TIME TO CLOSEST APPROACH [UTC]:	2047-11-03T14:22:09.882Z

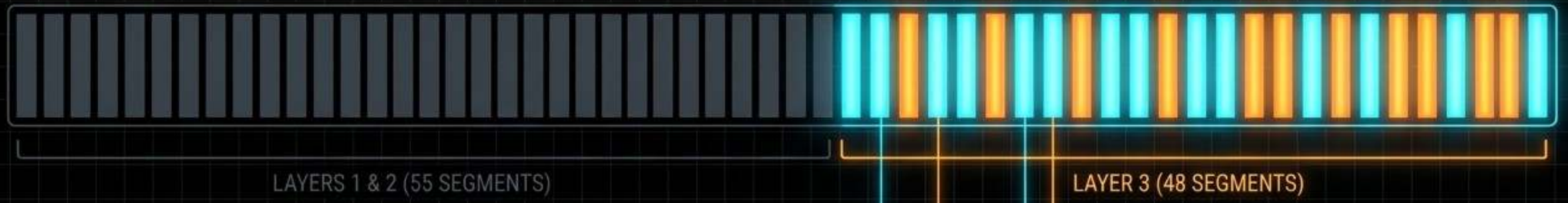
RISK METRICS

PROBABILITY OF COLLISION (Pc):	1.45e-04 (Crimson
MAHALANOBIS DISTANCE:	3.12σ (Cyan/Orange)
DATA DILUTION INDEX:	0.78

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[DECODING THE STATISTICAL BRAIN OF THE CDM]

While Layer 2 provides the raw physical miss distance, Layer 3 provides the context of uncertainty. A 100-meter miss distance is a blind, deceptive number without knowing the margins of radar error.



[THE 6x6 MATRICES]

(42 Columns)

Standard Deviations (σ) and Correlations (c).

[ELLIPSOID VOLUME]

(2 Columns)

Position Covariance Determinants.

[STATISTICAL DISTANCE]

(1 Column)

Mahalanobis Distance.

[THE RISK TRINITY]

(3 Columns)

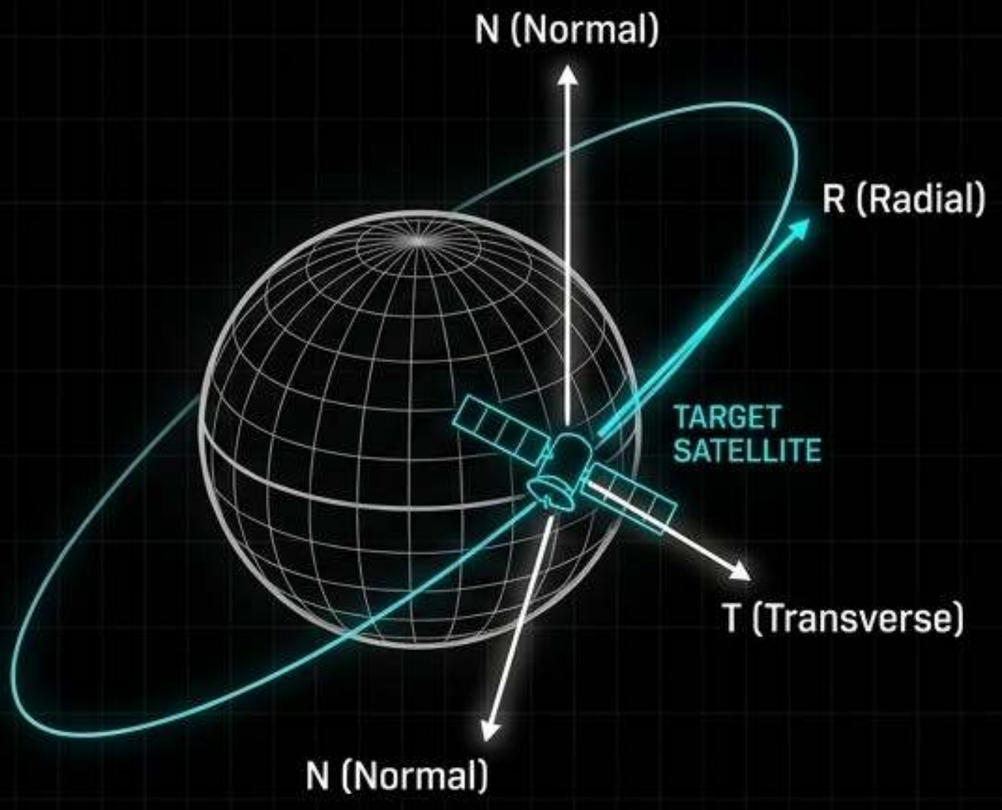
B-Plane Risk, Max Risk, and Risk Scaling.

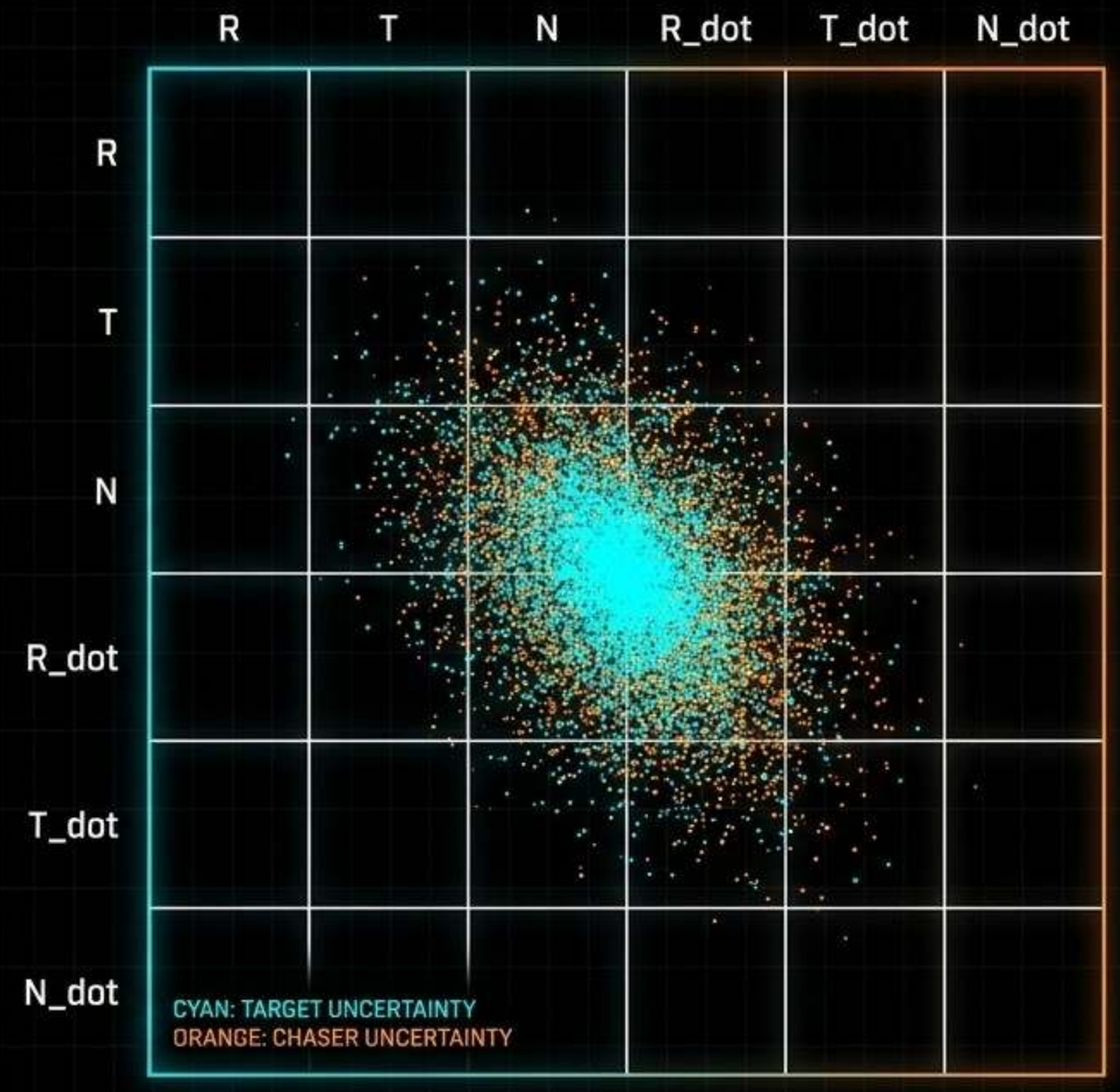
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[FROM STATE VECTOR TO UNCERTAINTY CLOUD]

At the Time of Closest Approach (TCA), the Target and Chaser each possess a nominal state vector in the RTN frame.

However, radar tracking is never perfect. The 6x6 Covariance Matrix captures the cloud of potential errors around this nominal point based on radar residuals and observation counts. The Target and Chaser each possess their own independent matrix.



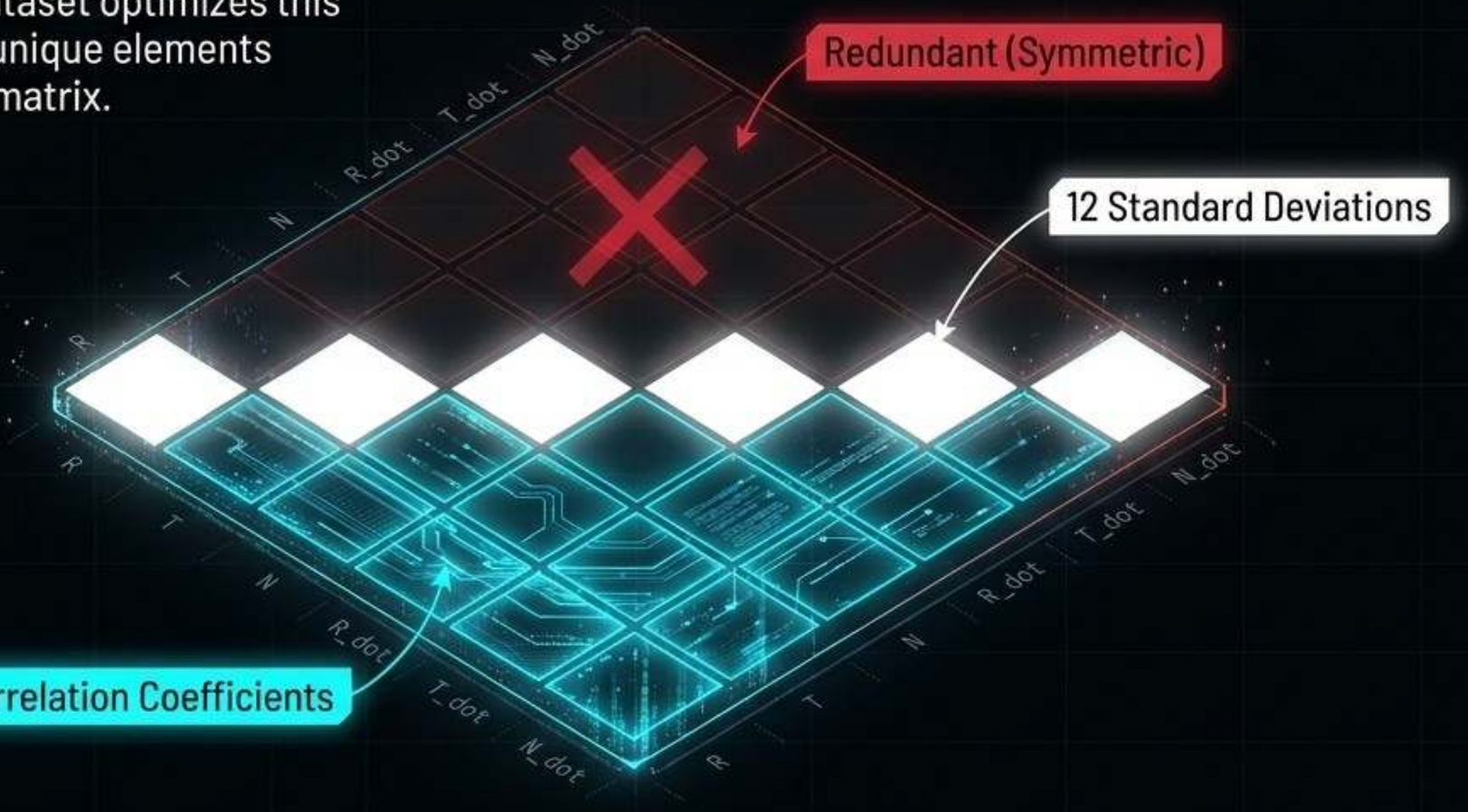


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HOW 72 CELLS BECOME 42 MACHINE LEARNING FEATURES

Because a covariance matrix is symmetric, the upper and lower triangles are identical. The ESA dataset optimizes this for ML models by extracting only the 21 unique elements from the lower triangle of each object's matrix.

- Target (t_): 6 Diagonals + 15 Off-Diagonals = 21 columns.
- Chaser (c_): 6 Diagonals + 15 Off-Diagonals = 21 columns.



(6 diagonal + 15 off-diagonal) × 2 objects = 42 ML columns

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[THE 12 STANDARD DEVIATIONS (σ): SIZING THE EGG]

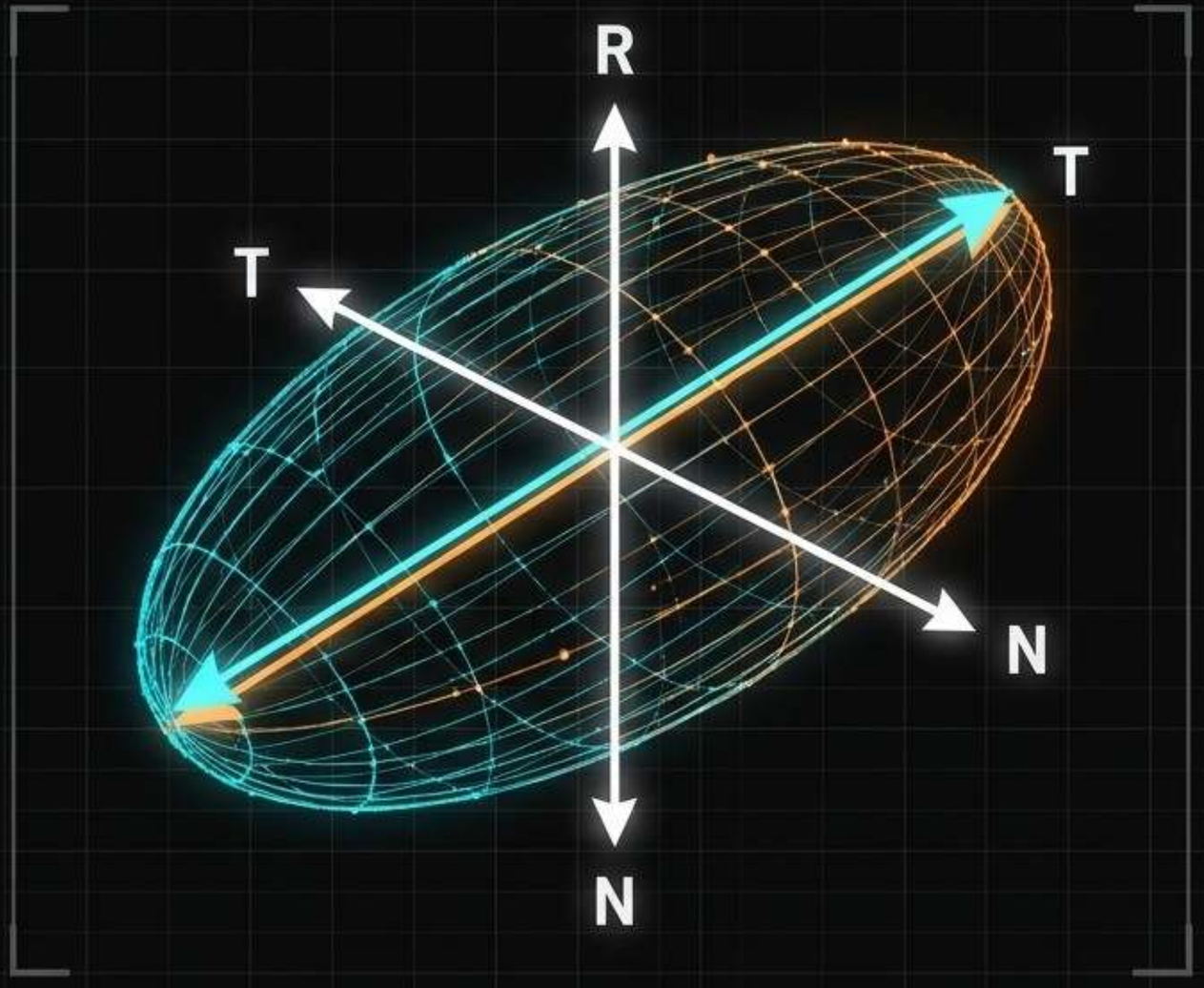
The main diagonal of the raw CDM matrix contains the Variance (m^2). To create the dataset features, the system takes the square root to extract the Standard Deviation (σ).

[Orbital Insight]

The Transverse/In-Track error (sigma_t) is almost always the largest. Uncertainty in atmospheric drag stretches the error ellipsoid into a massive cigar shape along the orbital path.

[Real Encounter Data (GALAXY 15 vs. Debris)]

Feature	Object	Value
t_sigma_t	Target Transverse Error	50.33 m
c_sigma_t	Chaser Transverse Error	1578.60 m

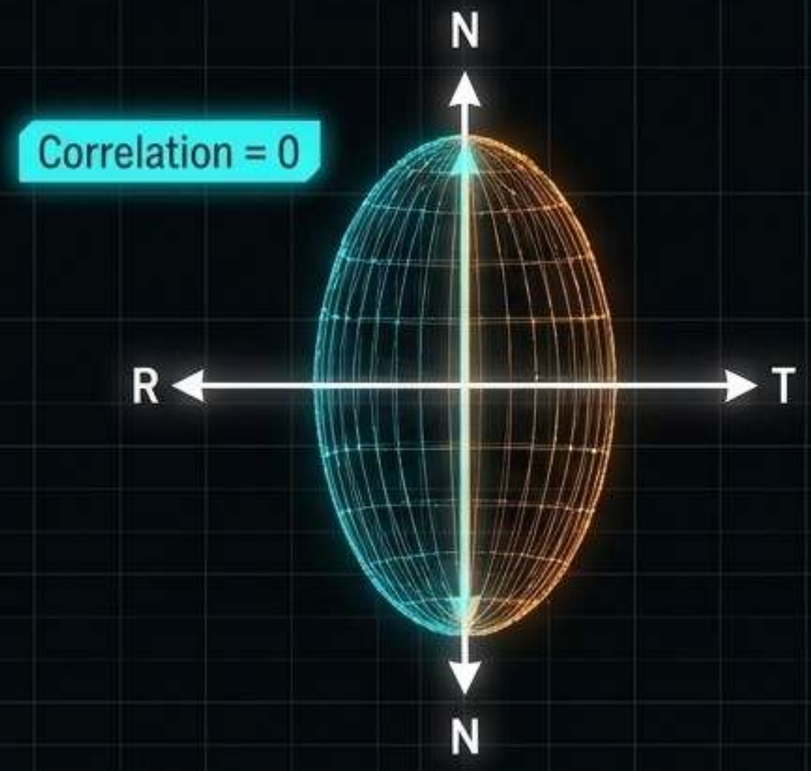


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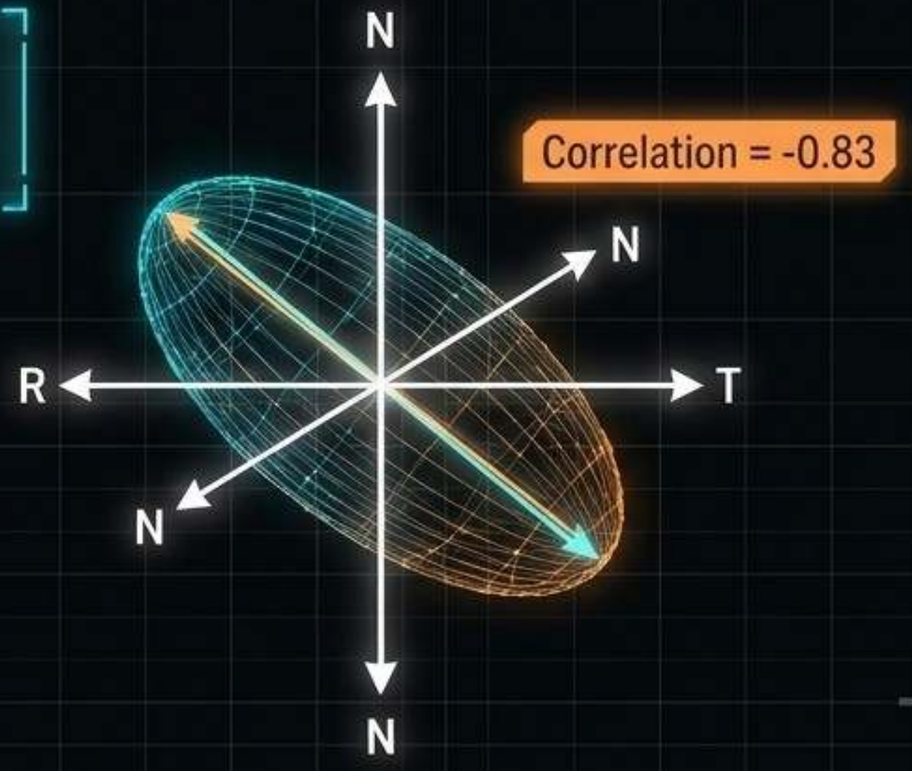
[THE 30 CORRELATION COEFFICIENTS (c): TILTING THE EGG]

The 15 lower-triangle elements are transformed from raw physical covariance into dimensionless Pearson Correlation Coefficients. These columns dictate the tilt and orientation of the error ellipsoid in 3D space.

Real Encounter Data: In our sample event, c_ct_r is -0.8327. This strong negative correlation indicates that an error in the Chaser's altitude (R) is fundamentally linked to an error in its track velocity (T), violently tilting its uncertainty cloud.



$$c_{ij} = \frac{c_{ij}}{(\sigma_i \cdot \sigma_j)}$$



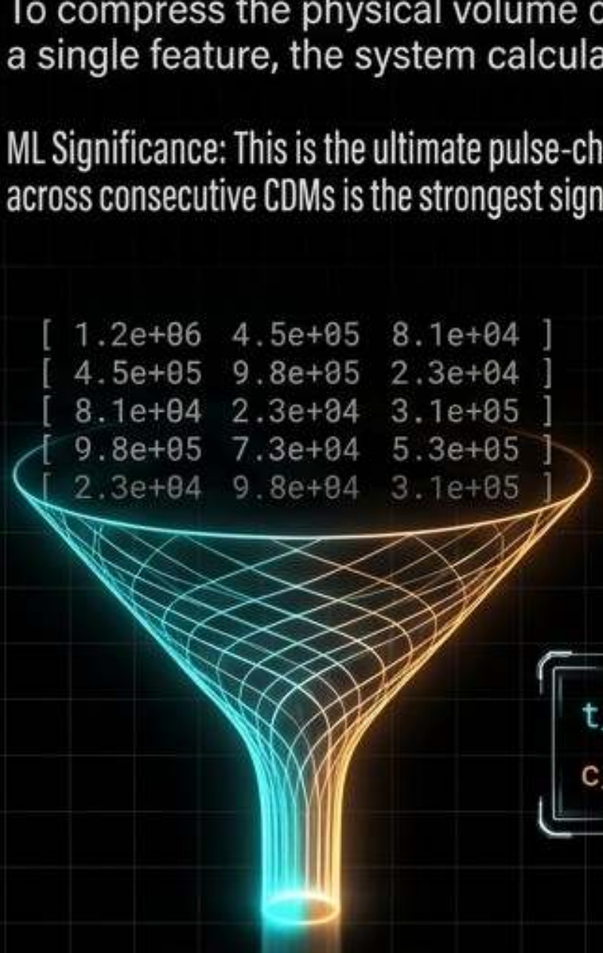
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[SUMMARIZING VOLUME: THE DETERMINANT (|C|)]

To compress the physical volume of the 3x3 Position Covariance sub-matrix into a single feature, the system calculates the Matrix Determinant.

ML Significance: This is the ultimate pulse-check for AI models. Tracking the rapid collapse of this determinant across consecutive CDMs is the strongest signal that new radar observations are locking onto the true trajectory.

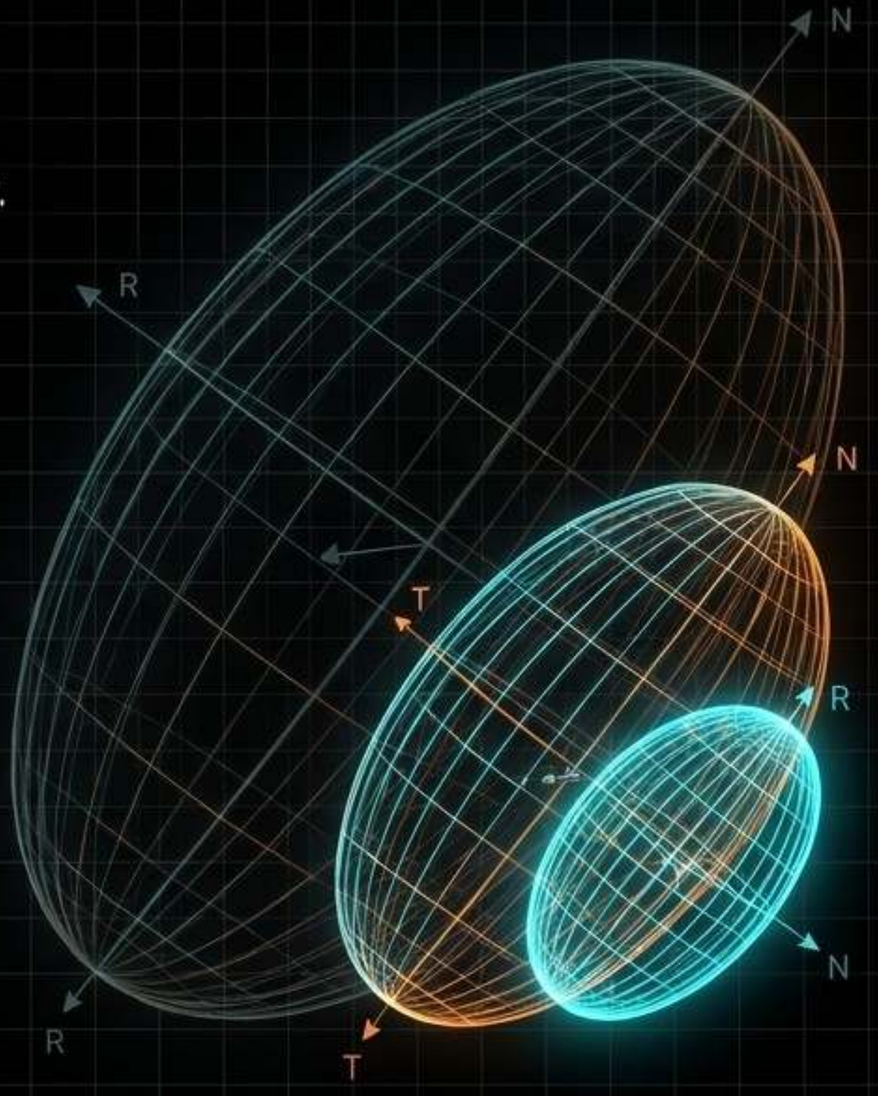
```
[ 1.2e+06  4.5e+05  8.1e+04 ]
[ 4.5e+05  9.8e+05  2.3e+04 ]
[ 8.1e+04  2.3e+04  3.1e+05 ]
[ 9.8e+05  7.3e+04  5.3e+05 ]
[ 2.3e+04  9.8e+04  3.1e+05 ]
```



[m^6]

t_position_covariance_det : 6.09×10^6 m^6

c_position_covariance_det : 7.24×10^{10} m^6



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[MERGING THE THREAT (C_comb)]

To calculate actual collision risk, we must establish a shared uncertainty space. We mathematically add the Target's 3×3 position covariance matrix to the Chaser's 3×3 position covariance matrix.

Context: This combined error ellipsoid now represents the total positional uncertainty of the encounter, setting the stage for evaluating the raw miss_distance.

$$\begin{bmatrix} c_11 & c_12 & c_13 \\ c_21 & c_22 & c_23 \\ c_31 & c_32 & c_33 \end{bmatrix} + \begin{bmatrix} c_11 & c_12 & c_13 \\ c_21 & c_22 & c_23 \\ c_31 & c_32 & c_33 \end{bmatrix} = \begin{bmatrix} c_11 & c_12 & c_13 \\ c_21 & c_22 & c_23 \\ c_31 & c_32 & c_33 \end{bmatrix}$$

```
C_comb = C_target + C_chaser
```



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[MEASURING DISTANCE IN SIGMAS (COLUMN 100)]

A physical miss_distance of 104m means nothing in isolation. Column 100 normalizes the relative position vector against the combined covariance matrix.

In our data, a 104.92m physical gap translates to a Mahalanobis distance of just 2.30σ. The satellite is dangerously deep inside the debris's statistical cloud.

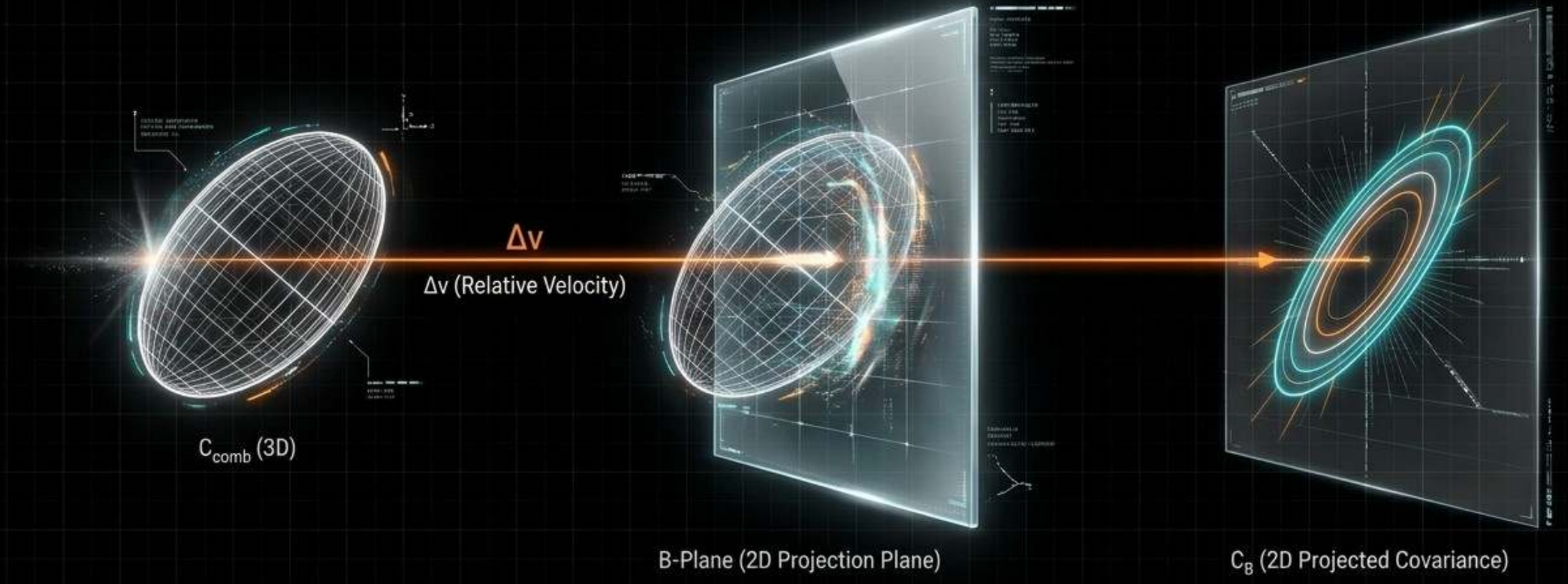
Formula: $d_M = \sqrt{\Delta r^T \cdot C_{\text{comb}}^{-1} \cdot \Delta r}$



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[PROJECTING THE ENCOUNTER (THE B-PLANE)]

Orbital encounters occur at ≈ 12 km/s. The crossing takes fractions of a millisecond—like a bullet piercing a sheet of paper. To calculate risk, ESA algorithms project the 3D combined covariance ellipsoid onto a 2D plane perpendicular to the relative velocity. The problem is instantly reduced to a solvable 2D Gaussian probability distribution.



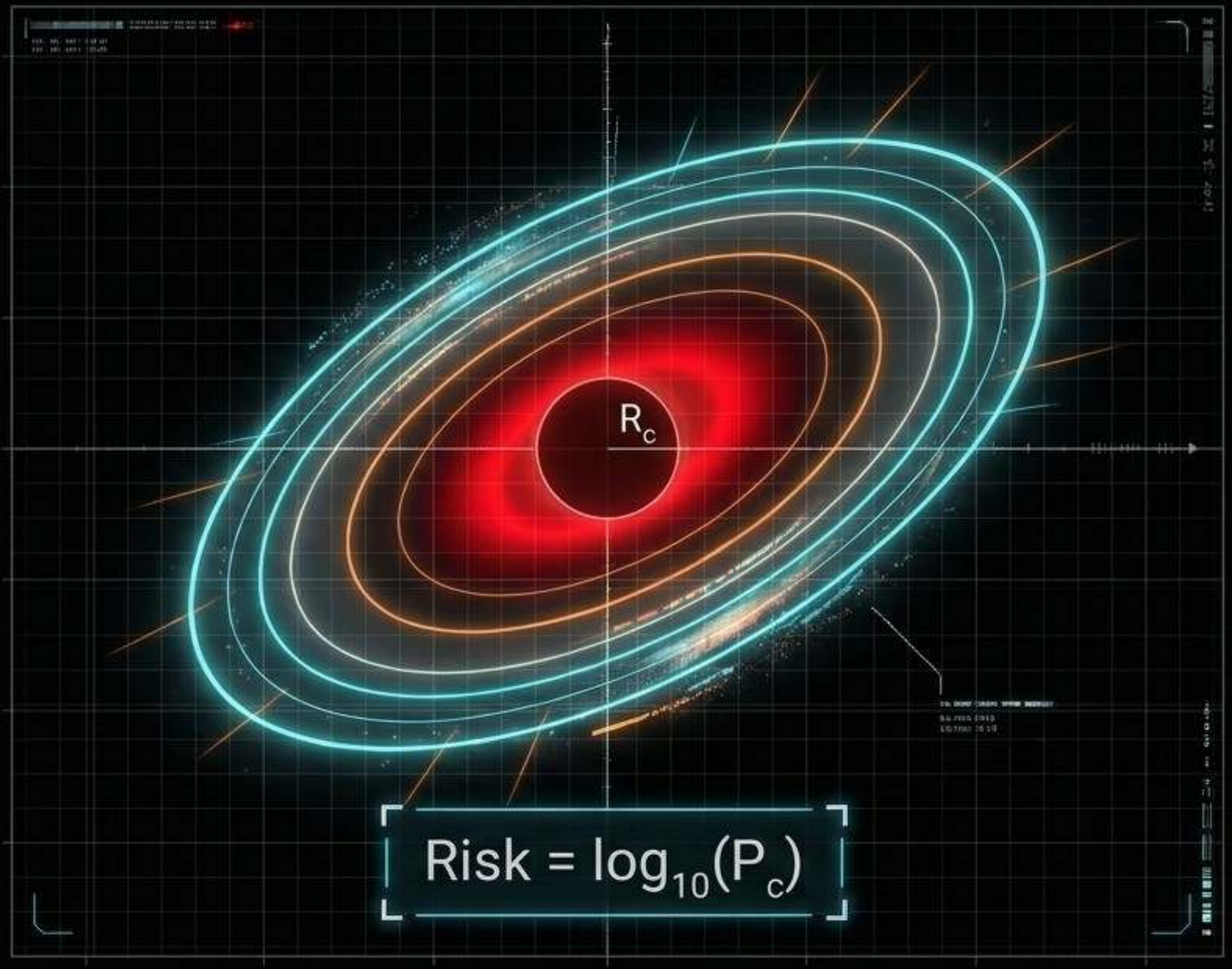
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[THE PROBABILITY OF COLLISION (P_c)]

Using the Alfriend-Akella algorithm, the system integrates the 2D Gaussian probability density function inside the 2.5m Hard-Body Radius circle. Because P_c yields tiny decimals, Column 101 converts it using a base-10 logarithm.

Real Encounter Data:
$P_c = 2.355 \times 10^{-3}$.
Therefore, risk = -2.628.

Thresholds: A risk of -6.0 is the critical baseline warning. A risk of -4.0 triggers a maneuver. This event (-2.628) is a severe red alert.



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[THE DEADLY ILLUSION OF LOW RISK]

The Data Dilution Paradox: If covariance is massive due to poor radar data, the probability integral is diluted over a vast volume of space, resulting in a deceptively low Risk score. As new radar data arrives and the ellipsoid naturally shrinks, the risk can suddenly spike into the red zone. Is a risk of -7.0 truly safe, or is the data just diluted?

[MASSIVE UNCERTAINTY = LOW RISK]



[DILUTED]

[FOCUSED UNCERTAINTY = HIGH RISK]



[PEAK THREAT]

[RESOLVED UNCERTAINTY = ZERO RISK]



[RESOLVED SAFETY]

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[FINDING THE WORST-CASE SCENARIO]

To solve Data Dilution, the algorithm mathematically tests all possible sizes of the covariance ellipsoid by scaling it by a factor of *k*.

Column 102 (max_risk_estimate):
The absolute highest collision probability possible if the ellipsoid was perfectly sized to maximize danger. (-2.387)

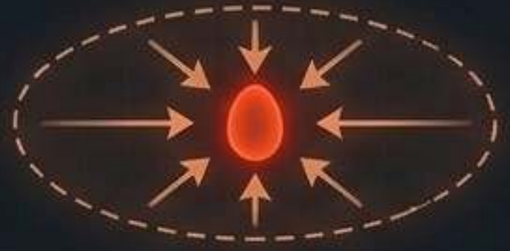
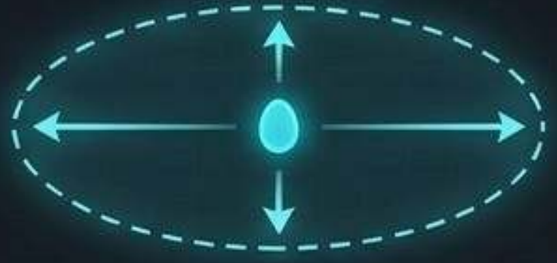
Column 103 (max_risk_scaling):
The exact multiplier (k) required to reach that maximum risk peak. (0.40)



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[DECODING THE SCALING FACTOR (k)]

[DIAGNOSTIC MATRIX]

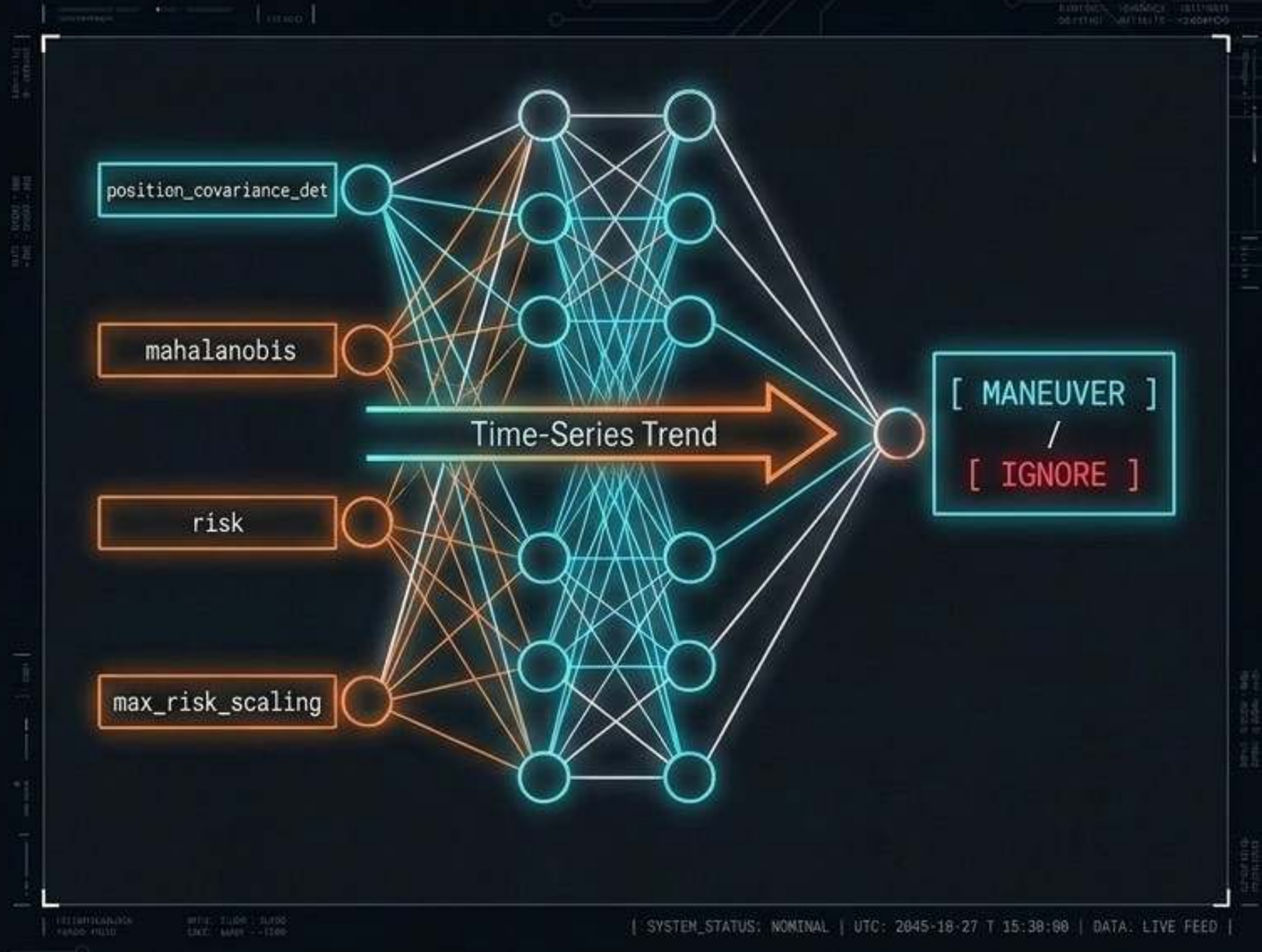
[$k \ll 1$ (e.g., 0.05)]		**DIAGNOSIS:** **Dilution Trap.** We must shrink the egg to 5% to reach max risk. The egg is currently bloated. Expect Risk to spike violently as TCA approaches.
[$k \approx 1$ (e.g., 1.0)]		**DIAGNOSIS:** **Peak Threat.** The egg is currently at the exact size to maximize danger. What you see is what you get.
[$k \gg 1$ (e.g., 20.0)]		**DIAGNOSIS:** **True Safety.** We must inflate the egg 20x to reach max risk. The object is small and far away.

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[LAYER 3 AS PREDICTIVE FEATURES]

A single CDM is just a snapshot. But by feeding a time-series of these 48 columns into models like LSTMs or Tree-Based Ensembles, AI learns to track the rate of ellipsoid collapse.

The Insight: The model learns to ignore diluted risk scores 3 days out, relying instead on `max_risk_scaling` and volumetric determinant trends. It accurately predicts the final P_c a full 48 hours before TCA. These 48 columns are not just geometry—they are the ultimate arbiters of orbital survival.



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